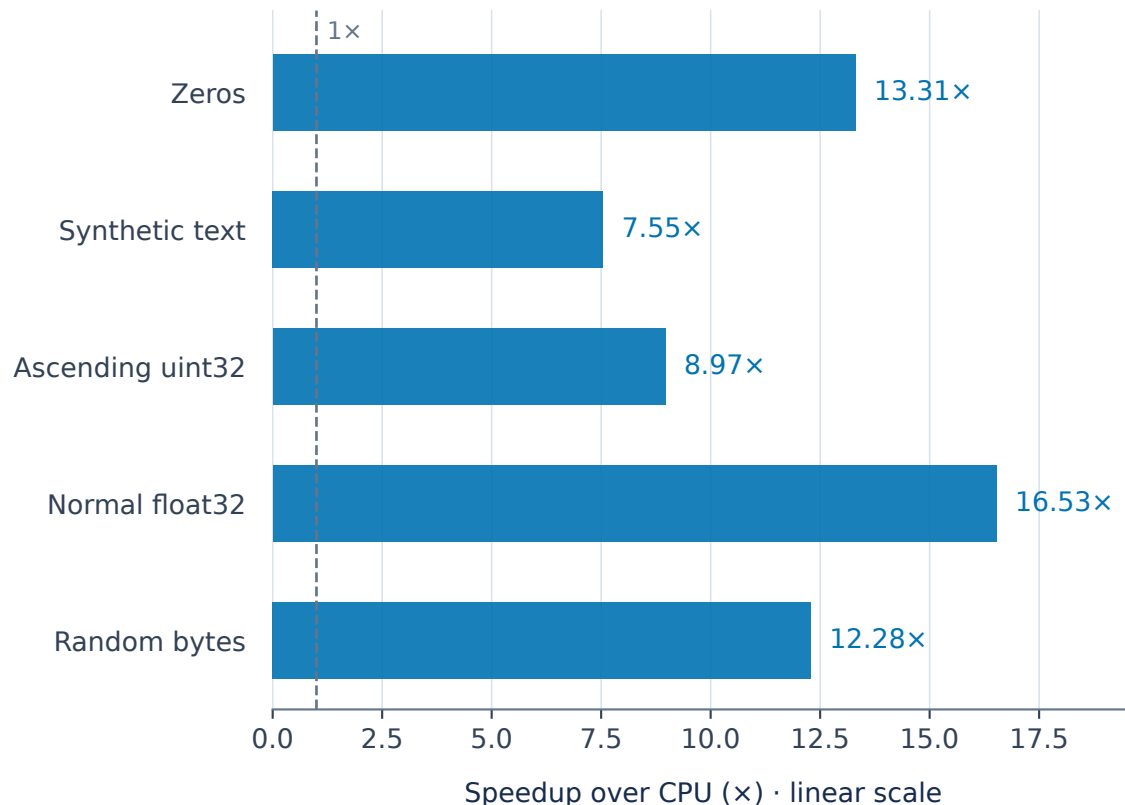


# GPU speedup over CPU

64 MiB inputs · CUDA host-to-host workflows return bytes and include copies and transfers

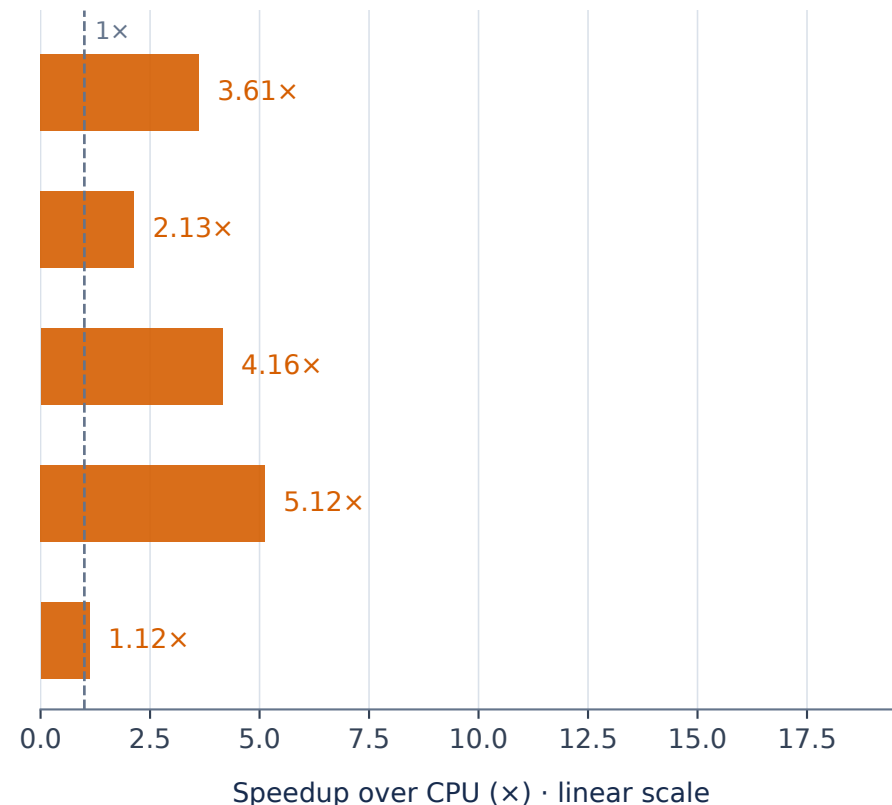
## Host-to-host compression

CPU: single-thread stdlib zlib level 1



## Host-to-host decompression

CPU: same stdlib level-6 bytes, single thread



RTX 4090 + Threadripper PRO 3995WX · same workstation · 2026-10-07

Ratio of warm CPU / CUDA median time; 15 CUDA / 5 CPU samples. Values above 1x favor CUDA. No error bars; cold startup excluded.

Compression compares each encoder's output with no level equivalence implied. Decompression uses identical level-6 streams.